HARISH MADURAIMANI S

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GitHub — Portfolio — LinkedIn

Profile

Electronics Engineer with a multidisciplinary mindset and a strong ability to learn, adapt and build solutions across hardware, software, AI and emerging technologies. Experienced in turning real world problems into practical intelligent systems by combining electronics, embedded systems, IoT, artificial intelligence, data science, robotics and software development. Passionate about exploring new technologies, connecting different engineering domains and developing scalable solutions from concept to prototype. Strong foundation in signal processing, analog and digital electronics, automation, Generative AI, PCB Design and VLSI, with an interest in system level design and hardwaresoftware co design. Driven by curiosity, innovation and the ability to take on diverse technical challenges beyond a single specialization.

Education

Bachelor's Degree in Electronics Engineering — **CGPA: 7.86/10**
Rajalakshmi Institute of Technology, Chennai — *Up to 4th Semester*

St. John Public School — **Class X(AISSE) – 2022 – 80%**

St. John Public School — **Class XII(AISSE) – 2024 – 80%**

Language

Tamil English Japanese (N5 Level Write.Read)

Internship & Experience

Kreotix Innovation — *Feb 2025 – Mar 2025*
- Gained practical exposure to technology development,PCB design,innovation and engineering problem solving.

Center of AI – Full AI System for Aviation — *2025*
- Worked on a full AI system for aviation, exploring intelligent decision support and AI driven aviation applications.

Authority of Air India — *2025*
- Gained exposure to aviation related systems,Communication Tech and operational engineering environments.

Projects

AI Copilot System — *AI, Predictive Models, Aviation*
- Developed an AI-based decision support system for aviation and Single Pilot Operations, integrating fuel, weather, and aircraft parameters for predictive route and fuel optimization.
- **Shortlisted for Boeing Innovation 2026** and completed an **IEEE research paper** based on the project.

3-Axis Robotic Arm Control Robotic Leg Operation — *MATLAB Simulink*
- Designed and simulated a 3-axis rotational robotic arm using mathematical joint dynamics and PID controllers, analyzing rise time, settling time, overshoot, and steady-state error.

Local Business Prediction AI System — *AI, Chatbot, Data Analysis*
- Developed an AI chatbot for location based business ideas with risk analysis, ROI prediction, demand mapping, openstreetmap and trend analysis; **shortlisted for the GDG Prefinal 2025**.

InfuSense — *IoT, Healthcare*
- Designed a **multi patient monitoring system** capable of tracking multiple IV drips simultaneously through a centralized interface, with potential application in government hospitals and healthcare facilities. **shortlisted for MSME** and represented the project at an IIT incubator event and a startup event.

SafeNest – Disaster-Resilient IoT Communication System — *ESP32, LoRa, GPS*
- Built an emergency communication system enabling long range transmission of GPS and biological data without cellular or internet connectivity using ACK based LoRa communication.

Mining Rescue Rover — *Robotics, Sensors, Wireless Communication*
- Developed an autonomous mining rover for exploration and monitoring in hazardous environments, integrating robotics, sensors, navigation and wireless communication.

Technical Skills

AI & ML: Computer Vision - YOLO, U-Net, LangChain - LCEL, LangGraph, RAG System, PyTorch, TensorFlow, Pandas, NumPy, LLM, VLM, Machine Learning, Deep Learning - CNN & RNN, NLP Models, Vector Database - pgvector, LangSmith, Edge AI, n8n

Embedded & IoT: Embedded C, STM32, 8051, Raspberry Pi 5, ESP32, IoT, Microcontrollers, FreeRTOS, Keil µVision, PlatformIO, Arduino IDE, Sensors, Wireless Communication, IoT Digital Twins, Blynk IoT, ThingSpeak, Adafruit IO, Node-RED, HiveMQ, ThingsBoard, Embedded Linux

Electronics & Hardware: Analog Electronics, Digital Electronics, Signal Processing - Scilab & Signal Analyzer (MATLAB), VLSI Design - Verilog and Microwind, LTspice, PCB Design (1 & 2 Layer) - KiCad, Altium

Robotics & Tools: ROS 2, Gazebo, MATLAB, Simulink, Control Systems - Scilab, Fusion 360 - Enclosure Design, Blender

Programming: Python, C, Rust (Basic), CUDA (Basic), Java, SQL, Web Development, Git & GitHub, UI/UX - Figma, Power BI, Google Analytics, Google Search Console

Certifications & Courses

NPTEL: Design Thinking-A Primer **(Certified)** Large Language Models;Python for Data Science;verilog(HDL) **(Courses)**
Simplilearn:VLSI Course,Introduction of IoT, Embedded System,PCB Design
MATWORKS: Matlab onramp,Signal Processing,Simulink,Control design
Research on Quantum Computing